Carrier Corporation • Syracuse, N.Y. 13221
SUBJECT TO CHANGE WITHOUT NOTICE
SHIPPING POINT: CHARLOTTE, NC
NEAREST PORT OF EMBARKATION: SAVANNAH, GA

TYPE OF SALE: 1634 300-500 (NA & Export)

F.O.B. SHIPPING POINT
FREIGHT PREPAID

30XF AQUAFORCE WITH GREENSPEED INTELLIGENCE AIR-COOLED ROTARY SCREW COMPRESSION LIQUID CHILLERS

These air-cooled packaged chiller units are fully piped and charged with HFC513A- refrigerant. Weatherproof construction, lightweight distribution and small footprint make them ideal for rooftop or ground level installation. Upward discharge airflow minimizes directional sound and carries heat away from surrounding areas (Notes 1, 2, 3, 4 & 5).

FEATURES AND ACCESSORIES

Refrigeration Circuit	30XF 300 – 500
Full R-513A Refrigerant Charge	Std
Full Load Factory Run Test and Run Test Report	Std
Independent Refrigeration Circuits	Std
Variable Speed Rotary Screw Compressors	Std
Compressor Protection (Note 6)	Std
~3600 Step Electronic Expansion Valve (EXV – Note 7)	Std
Two-Pass Flooded Shell-and-Tube Evaporator with Heater (Notes 8 and 12)	Std
Evaporator Insulation (Note 9)	Std
Filter Drier	Std
Moisture Indicating Sight Glass	Std
AeroAcoustic Condenser Fans	Std
Low-Sound Kit	FIOP
Micro Channel Condenser Coil	Std
Condenser Coil Options	
E-Coated Micro Channel	FIOP
Suction Service Valve Option	Std
Suction Line Insulation	Std
Liquid Service Valve per Circuit & Discharge Service Valve	Std
Waterside Economizer Coils with diverting valve	Std

PC 903 FORM NO. 30XF-2PPG
SUPERSEDES
PRICE SECTION 30XF

Chilled Water Circuit	30XF 300 - 500
150 psig (1034 kPa) Fluid Side Rated	Std
Electronic Flow Switch (thermal dispersion – Note 10)	Std
Freeze Protection (Note 11) – Evaporator Heaters (Note 12)	Std
Hydronic Kit	FIOP
Glycol Free Configuration	FIOP

Controls / Communication	30XF 300 - 500
Microprocessor 7-inch TOUCHPILOT (touch screen) Display	Std
Control Box Hinged Access Panels	Std
Temperature & Pressure Display	Std
Energy Management Module (EMM) + GFI Convenience OUtlet	Std
Chilled Water Temperature Reset (from return fluid – Note 13)	Std
Chilled Water Temperature Reset Sensor (via 4-20 mA signal; EMM required)	Std
Demand Limit, 1-step (via switch, field supplied)	Std
Demand Limit (via 4-20 mA signal, 2 additional steps; EMM required)	Std
Load Shedding Capability (BMS required, via CCN(std) or BACnet Communication)	Std
Low Water Temp Protection	Std
High Pressure Protection	Std
Low Pressure Protection	Std
Compressor Lead/Lag Control (cycles & run-time – Note 14)	Std
Unit Mounted Control Panel	Std
Variable Speed Condenser Fans	Std
Electronically Commutated Motor (ECM) for the condenser Fans	FIOP
Anti-Cycling Protection	Std
CCN Communication	Std
BACnet (IP MS/TP)	Std
~~LONworks Translator~~	~~Accy~~
External Pump Control Capability (no variable speed pump capability)	Std
Dual Chiller Accessory Kit	Accy
Display Heater for Control Panel	Accy
Datacenter Option	Std
Uninterrupted Power Supply (UPS)	Std
Compressor Boost	FIOP

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Electrical	30XF 300 - 500
Single Point Primary Power Entry	Std
Control Transformer (line power voltage to 115VAC)	Std
Unit Mounted Main Disconnect, Non-fused	Std
115V GFI Convenience Outlet (included on EMM option)	Std
Compressor Soft Start with VFD	Std
Permanent Magnet (PM) Compressor Motor	FIOP
SCCR – 25Ka	Std
High SCCR (65kA – 460v)	FIOP
Line Reactor	FIOP
Active Harmonic Filter	FIOP

Packaging / Security	30XF 300 - 500
Coil Face Shipping Protection	Std
Condenser Coil Trim Panels	FIOP / Accy
Full Hail Guard	FIOP

Std — Standard Feature
FIOP — Factory-Installed Option
Accy — Field-Installed Accessory
N/A — Not Available

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MODEL NUMBER NOMENCLATURE

Position	1-4	5	6-8	9	10	11	12	13	14	15	16	17
Description	**30XF**	**-**	**300**	**6**	**M**	**L**	**2**	**-**	**6**	**0**	**S**	**0**

Air-Cooled AquaForce Chiller

Design Series / Factory Location

Nominal Capacity

300 450
400 500

Voltage Options

6 = 460-3-60

Product Tiers & Evaporator Configuration

M = Medium
H = High

Condenser Options

Evaporator & Hydronic Kit Options

Low Ambient & Sound Options

Valve & Insulation Options

Electrical Options

Controls Options

Packaging Options

MODEL NUMBER NOMENCLATURE DEFINITION

Manufacturing Plant (Position 5)

- = Charlotte, NC - USA
M = Monterrey - Mexico

Voltage Options (Position 9)

6 = 460-3-60

Product Tier Configuration (Position 10)

M = Mid Tier
H = High Tier

Condenser and PM Motor Options (Position 11)

L = Micro Channel , R-513A Refrigerant
M = Micro Channel, E-Coat, R-513A Refrigerant
W = Micro Channel, PM Compressor Motor, R-513A Refrigerant
X = Micro Channel, E-Coat, PM Compressor Motor, R-513A Refrigerant

Evaporator and Hydronic Kit Options (Position 12)

2 = Flooded Evaporator, 2 pass for Brine Application, w/ Heater, no hydronic kit
8 = Flooded Evaporator, 2 pass for Brine Application, w/ Heater, hydronic kit low head pressure
9 = Flooded Evaporator, 2 pass for Brine Application, w/ Heater, hydronic kit high head pressure
Q = Flooded Evaporator, 2 pass for Brine Application, w/ Heater, no hydronic kit, glycol-free configuration
S = Flooded Evaporator, 2 pass for Brine Application, w/ Heater, hydronic kit low head pressure, glycol-free configuration
T = Flooded Evaporator, 2 pass for Brine Application, w/ Heater, hydronic kit high head pressure, glycol-free configuration

Condenser Fans and Sound Options (Position 13)

- = Variable Speed Fans (Mid and High-Tier), Std Sound
1 = Variable Speed Fans (Mid and High-Tier), Low Sound Kit
C = ECM Fans, Std Sound
D = ECM Fans, Low Sound Kit

Valve and Insulation Options (Position 14)

6 = Suction service valve, Actuated discharge valves (Brine evaporator), Suction line insulation, R-513A Refrigerant

Electrical Options (Position 15)

0 = Single Point Power w/ disconnect, Std SCCR
4 = Single Point Power w/ disconnect, Control transformer, Std SCCR
7 = Single Point Power w/ disconnect, High SCCR
B = Single Point Power w/ disconnect, Control transformer, High SCCR
F = Single Point Power w/ disconnect, Std SCCR, Active Harmonic Filter
G = Single Point Power w/ disconnect, Control transformer, Std SCCR, Active Harmonic Filter
J = Single Point Power w/ disconnect, High SCCR, Active Harmonic Filter
K = Single Point Power w/ disconnect, Control transformer, High SCCR, Active Harmonic Filter
M = Single Point Power w/ disconnect, Std SCCR, Line Reactor
N = Single Point Power w/ disconnect, Control transformer, Std SCCR, Line Reactor
Q = Single Point Power w/ disconnect, High SCCR, Line Reactor
R = Single Point Power w/ disconnect, Control transformer, High SCCR, Line Reactor

Control Options (Position 16)

S = 7” touchpilot, EMM, GFI convenience outlet, Remote Connectivity, Datacenter, Uninterruptible Power Supply (UPS)

Y = 7” touchpilot, EMM, GFI convenience outlet, Remote Connectivity, Datacenter, Uninterruptible Power Supply (UPS), Compressor Boost

Packaging Options (Position 17)

0 = Coil Face Shipping Protection (CFSP), Coil Trim Panels
3 = Full Hail Guards

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F.O.B. SHIPPING POINT
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30XF AIR COOLED LIQUID CHILLING PACKAGES
BASE UNIT WITH NO CRATING

ORDERING NUMBER	POWER SUPPLY	MASTER PRICE
30XF-3006ML2-60S0	460-3-60	934,523
30XF-3006HL2-60S0	460-3-60	1,002,900
30XF-4006ML2-60S0	460-3-60	1,043,692
30XF-4006HL2-60S0	460-3-60	1,112,019
30XF-4506ML2-60S0	460-3-60	1,114,784
30XF-4506HL2-60S0	460-3-60	1,143,561
30XF-5006ML2-60S0	460-3-60	1,156,032
30XF-5006HL2-60S0	460-3-60	1,380,369

FACTORY-INSTALLED OPTIONS (FIOP)

POSITION 11

CONDENSER COIL CORROSION PROTECTION OPTION

DESCRIPTION	MASTER PRICE (Pricing done based on the # of V's)					
	8V	9V	10V	11V	12V	13V
	Where used	Where used	Where used	Where used	Where used	Where used
	300-Mid	300-High	400-Mid	400-High 450-Mid	450-High 500-Mid	500-High
Micro Channel	Std.	Std.	Std.	Std.	Std.	Std.
Micro Channel, E- Coat X M **1N**	30XF_COIL_002 41,250	30XF_COIL_003 46,406	30XF_COIL_004 51,564	30XF_COIL_001 56,720	30XF_COIL_005 61,880	30XF_COIL_006 67,036

PERMANENT MAGNET COMPRESSOR MOTOR

DESCRIPTION	MASTER PRICE		
	Voltage	300	400-500
No PM Motor	All	Std.	Std.
Permanent Magnet Compressor Motor W X **17A**	460-3-60 6	30XF_COMPR_002 18,288	30XF_COMPR_001 18,529

POSITION 12

EVAPORATOR OPTION

DESCRIPTION	MASTER PRICE			
	300-Mid	**300-High**	**400-Mid 400-High**	**450-Mid 450-High 500-Mid**
Flooded Evaporator, 2 pass, with heater for Brine Application	Std.	Std.	Std.	Std.

HYDRONIC KIT OPTION*

DESCRIPTION	MASTER PRICE				
	Voltage	**300-High**	**400-Mid**	**400-High 450-Mid**	**450-High 500-Mid 500-High**
No Pump	460	Std	Std.	Std.	Std.
Low Head Pump (N+0) 8 S **16A 16C**	460	30XF_PUMP_001 99,982	30XF_PUMP_003 99,982	30XF_PUMP_005 99,982	30XF_PUMP_007 99,982
High Head Pump (N+0) 9 T **16B 16D**	460	30XF_PUMP_002 99,982	30XF_PUMP_004 99,982	30XF_PUMP_006 99,982	30XF_PUMP_008 99,982

*Pumps not available on 300M and 350M

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GLYCOL-FREE OPTION

DESCRIPTION	MASTER PRICE				
	300-Mid	**300-High**	**400-Mid*** **400-High**	**450-Mid** **450-High** **500-Mid**	**500-High**
No Glycol-free	Std.	Std.	Std.	Std.	Std.
Glycol-free *(not available for 30XF400M)	30XF_GLYCOL_0011	30XF_GLYCOL_0021	30XF_GLYCOL_003 233,315	30XF_GLYCOL_004 233,315	30XF_GLYCOL_005 233,315

POSITION 13

SOUND OPTIONS

DESCRIPTION	MASTER PRICE	
	300	400-500
Standard Sound - No compressor sound enclosures	Std.	Std.
Low Sound Kit - Includes full compressor sound enclosure and muffler for each compressor 1 D **4A**	30XF_SOUND_001 20,954	30XF_SOUND_002 20,954

CONDENSER FANS

DESCRIPTION	MASTER PRICE					
	Voltage	300-Mid 300-High	400-Mid	400-High 450-Mid	450-High 500-Mid	500-High
Variable Speed Condenser Fans*	460	Std	Std.	Std.	Std.	Std.
Electronically Commuted Motor (ECM) Fans * ‡ **C D 3C**	460	30XF_CONDFAN_002 23,708	30XF_CONDFAN_003 39,747	30XF_CONDFAN_001 44,416	30XF_CONDFAN_004 49,682	30XF_CONDFAN_005 53,822

* (includes providing the function of Low Ambient Head Pressure Control) - This option permits start-up at 0°F (-18°C) and operation of the 30XF units to -20°F (-29°C) outdoor ambient temperature by controlling the speed of the condenser fans. Package includes sensor, control and hardware for operating the fans in each circuit. With pump control or glycol in loop; without pump control or glycol, minimum operating ambient temperature is 0° F

‡ May require installation of field-supplied wind baffles (not included)

FACTORY-INSTALLED OPTIONS (cont)

POSTION 14

ISOLATION VALVE

DESCRIPTION	MASTER PRICE	
	300	400-500
No Additional Isolation Valve(s) (Std)	-	-
Isolation Valve(s) - Provides additional refrigerant isolation capability. Regardless of which cooler option is employed, the selection of the isolation valve option results in chillers which are equipped with a liquid line service valve, a discharge service valve, and a series of valves on or near the cooler, the net effect of which is to provide isolation capability in the condenser area, the cooler area and the compressor area. Please refer to the latest PD document for additional details.	Std	Std

SUCTION LINE INSULATION OPTION

DESCRIPTION	MASTER PRICE	
	300	400-500
No Suction Line Insulation (Std)	-	-
Suction Line Insulation - Insulation is tubular closed-cell insulation. This option is recommended for areas of high dew points where condensation may be a concern. *	Std	Std

* Suction line insulation is valid for any operating condition.

POSITION 15

ELECTRICAL OPTIONS

CONTROL TRANSFORMER OPTION

DESCRIPTION	Voltage	MASTER PRICE
		300-500
None	460-3-60	Std.
Control Transformer - Main power source supplies control circuit (line power voltage to 115VAC) **9A**	460-3-60	30XF_CNTLTF_001 4,097

POWER CONNECTION OPTION

DESCRIPTION	VOLTAGE	MASTER PRICE			
		300	400	450	500
Single Point	460-3-60	Std.	Std.	Std.	Std.

SHORT CIRCUIT CURRENT RATING (SCCR) OPTION

DESCRIPTION			UNIT SIZE (Tier)				
Power Connection	SCCR	Description	Voltage	SCCR RATING	300	400	450-500
Single Point	Standard	Disconnect Switch	460-3-60	25kA	Std.	Std.	Std.
Single Point 0 7 B F G J K M N Q R **8A**	High 7 B J K Q R **10A**	* Disconnect and breakers	460-3-60 **6**	65kA	30XF_SCCR_001 7,611	30XF_SCCR_003 7,611	30XF_SCCR_005 7,611

ACTIVE HARMONIC FILTER*

DESCRIPTION	VOLTAGE	MASTER PRICE			
		300	400	450	500
No Harmonic Filter	460-3-60	Std.	Std.	Std.	Std.
Active Harmonic Filter – provides IEEE-519 compliance F G J K **22A**	460-3-60	30XF_FILTER_002 172,442	30XF_FILTER_001 172,442	30XF_FILTER_003 172,442	30XF_FILTER_004 172,442

* AHF only available with single point power. Not available with Line Reactor

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LINE REACTOR*

DESCRIPTION	VOLTAGE	MASTER PRICE			
		300	400	450	500
No Line Reactor	460-3-60	Std.	Std.	Std.	Std.
Line Reactor M N Q R **22B**	460-3-60	30XF_REACTOR_001 1	30XF_REACTOR_002 3829	30XF_REACTOR_003 3829	30XF_REACTOR_004 3829

* AHF only available with single point power. Not available with Harmonic Filter

POSITION 16

CONTROLS OPTIONS

ENERGY MANAGEMENT MODULE

DESCRIPTION	UNIT	MASTER PRICE
Energy Management Module (EMM) - includes GFI convenience outlet	All	Std

EXTERNAL INTERFACE OPTION

DESCRIPTION	VOLTAGE	MASTER PRICE	
		300	400-500
BACnet / Modbus Translator (Building automation system over BACnet or Modbus)	All	Std.	Std.

DATACENTER OPTION

DESCRIPTION	VOLTAGE	MASTER PRICE
		All Sizes
Datacenter Option: Allows chiller to operate with higher Entry and Leaving Water Temperatures at Higher Ambient Conditions	460	Std.

UNINTERRUPTIBLE POWER SUPPLY (UPS)

DESCRIPTION	VOLTAGE	MASTER PRICE			
		300	400	450	500
No Uninterruptible Power Supply	460-3-60	-	-	-	-
Uninterruptible Power Supply (UPS)	460-3-60	Std.	Std.	Std.	Std.

COMPRESSOR BOOST

DESCRIPTION	MASTER PRICE			
	300	400	450	500
No Compressor Boost	Std.	Std.	Std.	Std.
Compressor Boost – Improves overall chiller efficiency	30XF_CMPBOOST_001 1	30XF_CMPBOOST_002 3000	30XF_CMPBOOST_003 3000	30XF_CMPBOOST_004 3000

POSITION 17

SECURITY GRILLES AND HAIL GUARD OPTIONS

DESCRIPTION	MASTER PRICE (Pricing done based on the # of V's)					
	8V	**9V**	**10V**	**11V**	**12V**	**13V**
	Where used	Where used	Where used	Where used	Where used	Where used
	300-Mid	300-High	400-Mid	400-High 450-Mid	450-High 500-Mid	500-High
Coil Trim Panels*	Std.	Std.	Std.	Std.	Std.	Std.
Full Hail Guards 3 **14D**	30XF_PKG_002 14,050	30XF_PKG_003 15,205	30XF_PKG_001 16,882	30XF_PKG_004 16,854	30XF_PKG_005 17,808	30XF_PKG_006 18,762

***** Coil Trim Panels option will include the panels only for the header side

FIELD-INSTALLED ACCESSORIES

CONTROL/COMMUNICATION ACCESSORIES

ORDERING NUMBER	DESCRIPTION	USAGE	MASTER PRICE	RCD LIST
30XV70000701	Energy Management Module Kit - Where External temperature reset is required by 4-20mA signal. Package also provides Demand Limiting and temperature set point.	All	1,696	
30XV70001701	BACnet/Modbus Translator	All	924	
30XV70001901	LonWorks Translator	All	1,185	
33ZCT55SPT	Space Temp. w/ Occupancy Override Button	All	¤	
HH79NZ014	Thermistor (5k Ohm) - 3 inch Water Temperature Sensor	All	65	*
HH79NZ029	Thermistor (5k Ohm) - 4 inch Water Temperature Sensor	All	90	*
10HB50106801	Thermistor Well - 3 inch	All	58	*
10HB50106802	Thermistor Well - 4 inch	All	53	*
00EFN900044000A	Dual Chiller Accessory Kit - Provides the additional hardware required to operate two chillers in parallel.	All	443	
30XV70001601	Display Heater for Control Panel	All	758	

* RCD List Price - Subject to change on a quarterly basis and should be verified with RCD prior to quoting.
¤ See CCN System and Controls price page (catalog 500-059) or check SAP for current pricing.

ELECTRICAL ACCESSORIES

ORDERING NUMBER	DESCRIPTION	UNIT	SHIP WT (Lbs/Kgs)	MASTER PRICE
30XV70000801	GFI Convenience Outlet - includes a 4-amp GFI (Ground Fault Interrupt) receptacle with independent fuse protection. Convenience outlet is 115V female receptacle. Not available with 380-400V & all 50Hz.	All	10/4.5	1,318

COIL TRIM PANEL ACCESSORIES

For units in the field **with no Coil Trim Panels** FIOP and customer requires coil trim panels on both sides, the following must be ordered:

ORDERING NUMBER	DESCRIPTION	MASTER PRICE	MASTER PRICE (Pricing done based on the # of V's) Where used and Quantity					
			8V	9V	10V	11V	12V	12V
			30XF300M	30XF300H	30XF400M	30XF400H 30XF450M	30XF450H 30XF500M	30XF500H
30XV70000101	End Coils (2Vs for the ends + 2 center cover panels; 4 total Vs in this option) - mounting hardware included. Provides aesthetic, finished appearance for the condenser coil ends and the evaporator/pump connection side of the unit. This item is sufficient for a 4V unit	4,873	1	1	1	1	1	1
30XV70000201	Center Coils - 1V Module Kit (Center Cover Panels - one for each side of unit) For units with more than 4Vs, this module needs to be ordered as well. The quantity will depend on the amount of Vs the unit may have.	1,838	4	5	6	7	8	9

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***FACTORY START UP & FIRST YEAR LABOR (USA & CANADA)**

Order No.	DESCRIPTION	USAGE	NET PRICE
30XV-ST1	Start up, First Unit	All 30XF Models	2,628
30XV-STA	Start up, Each Additional Unit	All 30XF Models	1,772

***Required for all 30XF sales in USA and Canada**

GENERAL & APPLICATION NOTES

1. The units are designed for outdoor applications and, therefore, all electrical components are protected from the weather. All external sheet metal parts are painted with polyester baked enamel on galvanized steel. The frame is constructed of heavy-gage galvanized steel members zinc-dipped for superior corrosion protection.
2. All chillers are certified in accordance with the AHRI Air-Cooled Water-Chilling Packages Certification program, which is based on AHRI Standard 550/590 (I-P) and AHRI Standard 551/591 (SI). Certified units may be found in the AHRI Directory at www.ahridirectory.org.
3. All 460V 60 Hz are ETL and ETLc approved.
4. International Sales: F.O.B. Shipping Point. Freight prepaid to nearest point of embarkation.
5. Warranty Terms: APO, LAO, ETO Sales. Warranty terms and conditions determined by individual Carrier International Selling Company.
6. The compressors are electrically and mechanically protected by the VFDs and unit controls. Example of protection: electrical overload, over temperature, high pressure, low pressure/loss of charge, phase loss, phase reversal, etc.
7. The electronic expansion valve contains a linear stepper motor (~3600 steps) controlled by the unit's microprocessor to accurately meter refrigerant to the evaporator to maintain optimal operation of the system at all times
8. The evaporators, oil separators and mufflers ordered are designed, constructed and tested in compliance with the ASME Code requirements for Unfired Pressure Vessels. The ASME stamping (U) is applied to the refrigerant side only.
9. Evaporators are insulated with ¾-inch closed-cell foam.
10. A factory-installed electronic thermal dispersion flow switch provides chilled water proof-of-flow.
11. Low temperature override is provided to prevent overshoot from the leaving water that could cause the unit to go out on evaporator freeze protection.
12. Insertion evaporator heaters are standard on flooded evaporators. Whenever the unit is subject to temperatures below 33º F (1º C), an appropriate antifreeze solution must be added to the water in the evaporator to prevent freeze-up. The antifreeze should be properly inhibited to minimize corrosion.
13. Temperature reset from return, space temperature or outside air temperature may be performed with standard unit controls (requires accessory sensors). Reset via 4-20 mA signal requires the Energy Management Module.
14. Automatic lead/lag feature alternates lead compressor based on start cycles or run-time to ensure even wear for multiple compressor units.
15. Under many operating conditions, unit operation to as low as -20 F (-29 C) ambient temperature is achievable. For any operation below 32 F (0 C), pump control or the use of glycol in the system is required. Whenever a unit is expected to operate at ambient temperatures below 32 F (0 C), field-supplied and installed wind baffles must be added to the unit.

ETLc

ISO 9001

AHRI 550/590

ASME

OPTIONAL EXTENDED WARRANTIES ARE AVAILABLE. SEE CATALOG NO. 500-089 "COMMERCIAL START-UP AND OPTIONAL EXTENDED WARRANTY PRICING"